

Problem 15.12

A loan of \$20,000 to purchase a car at annual rate of interest of 6% will be paid back through monthly installments over 5 years, with 1st installment to be made 1 month after the release of the loan. What is the monthly installment?

Solution

The loan is repaid with monthly installments for 5 years, so the number of payments is

$$n = 5(12) = 60.$$

The annual effective interest rate is 6%. Let j be the monthly effective interest rate. Then

$$1.06 = (1 + j)^{12}.$$

Therefore,

$$j = (1.06)^{1/12} - 1.$$

Thus,

$$j \approx 0.00486755.$$

Equivalently, the nominal rate convertible monthly is

$$i^{(12)} = 12j \approx 0.0584107.$$

Let X be the monthly installment. Since the first payment is made one month after the loan is released, this is an annuity-immediate. Hence,

$$20000 = Xa_{\overline{60}|j}.$$

Using the present value formula at the monthly effective rate j ,

$$a_{\overline{60}|j} = \frac{1 - (1 + j)^{-60}}{j}.$$

Therefore,

$$X = \frac{20000}{\frac{1 - (1 + j)^{-60}}{j}}.$$

Substituting $j = (1.06)^{1/12} - 1$,

$$X = \frac{20000}{\frac{1 - (1 + (1.06)^{1/12} - 1)^{-60}}{(1.06)^{1/12} - 1}}.$$

Finally,

$$X \approx 385.179661.$$

Hence, the monthly installment is approximately

$$\boxed{\$385.18}.$$